

13-2b From the Production Function to the Total-Cost Curve

Columns (4), (5), and (6) in [Table 1](#) show Chloe's cost of producing cookies. In this example, the cost of Chloe's factory is **\$30** per hour, and the cost of a worker is **\$10** per hour. If she hires **1** worker, her total cost is **\$40** per hour. If she hires **2** workers, her total cost is **\$50** per hour, and so on. With this information, the table now shows how the number of workers Chloe hires is related to the quantity of cookies she produces and to her total cost of production.

Our goal in the next several chapters is to study firms' production and pricing decisions. For this purpose, the most important relationship in [Table 1](#) is between quantity produced [in column (2)] and total cost [in column (6)]. Panel (b) of [Figure 2](#) graphs these two columns of data with quantity produced on the horizontal axis and total cost on the vertical axis. This graph is called the *total-cost curve*.

Now compare the total-cost curve in panel (b) with the production function in panel (a). These two curves are opposite sides of the same coin. The total-cost curve gets steeper as the amount produced rises, whereas the production function gets flatter as production rises. These changes in slope occur for the same reason. High production of cookies means that Chloe's kitchen is crowded with many workers. Because the kitchen is crowded, each additional worker adds less to production, reflecting diminishing marginal product. Therefore, the production function is relatively flat. But now turn this logic around: When the kitchen is crowded, producing an additional cookie requires a lot of additional labor and is thus very costly. Therefore, when the quantity produced is large, the total-cost curve is relatively steep.

QuickQuiz

3. Farmer Greene faces diminishing marginal product. If she plants no seeds on her farm, she gets no harvest. If she plants **1** bag of seeds, she gets **3** bushels of wheat. If she plants **2** bags, she gets **5** bushels. If she plants **3** bags, she gets
- a. **6** bushels.
 - b. **7** bushels.
 - c. **8** bushels.
 - d. **9** bushels.

4. Diminishing marginal product explains why, as a firm's output increases,
- a. the production function and total-cost curve both get steeper.
 - b. the production function and total-cost curve both get flatter.
 - c. the production function gets steeper, while the total-cost curve gets flatter.
 - d. the production function gets flatter, while the total-cost curve gets steeper.

Chapter 13: The Costs of Production: 13-2b From the Production Function to the Total-Cost Curve

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